03: Inference for SLR

Stat 230 - Spring 2026

1 Part 1: On paper[1]

Biologists know that the leaves on plants tend to get smaller as temperatures rise, and average temperatures have been getting hotter over time. A sample of 252 leaves from the species *Dodonaea viscosa* subspecies *angustissima* (broadleaf hopbush) have been collected in a certain region of South Australia. Is there an association between leaf width and year? Below is the coefficient table from a simple linear regression model.

term	estimate	std.error	statistic	p.value
(Intercept)	37.723	8.575	4.399	<0.001
Year	-0.018	0.004	-4.029	<0.001

Table 1: lm output

Note: The width is the average width, in mm, of leaves, taken at their widest points, that were collected in a given year.

a. Report the fitted regression equation.

b. Interpret the slope of the regression line in context of the data.

[1] Adapted from *Stat 2: Modeling with Regression and ANOVA*

c. Calculate a 95% confidence interval for the slope of the regression line. (Use the slides to find the appropriate critical value). Interpret this interval in context of the data.

d. What hypotheses are being tested with the statistic and associated p-value in the `(Intercept)` row of the table? What conclusion can you draw?